

Solution Requirements

Date	15 March 2026
Team ID	XXXXXX
Project Name	Smart Lender
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	The system provides secure user access through the web application. If user authentication is implemented, users must log in using valid credentials before accessing prediction features.	High
2	Authorization Levels	The system supports role-based access. Administrators can manage the application and model, while end users can submit loan details and view prediction results.	Medium
3	External Interfaces	The system interacts with a trained Machine Learning model (.pkl), HTML/CSS frontend, Flask backend, and Python libraries such as Pandas, NumPy, and Scikit-learn. Future versions can integrate banking APIs and credit score services.	High
4	Transaction Processing	The system accepts applicant details, validates the inputs, preprocesses the data, passes it to the trained model, and returns the loan approval prediction without modifying the original dataset.	High
5	Reporting	The application displays the loan prediction result to the user. Model evaluation reports, including accuracy, confusion matrix,	Medium

S.No	Requirement Category	Requirement Description	Priority
		classification report, and cross-validation scores, are generated during development for performance analysis.	
6	Business Rules	Loan approval is determined based on applicant attributes such as income, credit history, employment status, education, loan amount, loan term, and property area. Input values must satisfy validation rules before prediction.	High
7	Compliance to Laws or Regulations	The system is designed to protect user information and follows standard data privacy principles. Personal information should be handled securely and used only for prediction purposes.	Medium
8	Other	The application provides a simple and user-friendly interface, supports fast prediction, and allows easy deployment using the Flask framework.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The system should process user input and generate loan predictions quickly with minimal delay.	Prediction result displayed within 2 seconds .
2	Scalability	The application should support increasing numbers of users and larger datasets without affecting performance.	Supports multiple concurrent users and can be extended for enterprise deployment.
3	Security & Data Privacy	Applicant information must be processed securely. Input validation should prevent invalid or malicious data, and the trained model should be protected from unauthorized access.	Secure data handling, input validation, and restricted model access.
4	Reliability & Availability	The application should provide consistent predictions and remain available during normal operating conditions.	99% application availability with reliable prediction results.
5	Usability & Accessibility	The user interface should be simple, responsive, and easy to navigate for users with basic computer knowledge.	Users should be able to complete a prediction in less than 3 minutes with minimal guidance.
6	Maintainability & Portability	The application should be modular, easy to maintain, and deployable on Windows, Linux, or cloud platforms with minimal configuration changes.	Modular code structure with support for cross-platform deployment.

